

Project Assingment 1 & 2 (P1P2)

Team #22

Team members:

Bader AlKhawashki, Arunima Basu, Fritz Kuermayr, Aman Shah, Salvador Gonzalez

Deliverable 1: Team Project Agreement (for future reference)

1) Shop training:

Fritz Kuermayr will undergo the shop training. Aman Shah & Salvador Gonzalez will assist in the hardware, software, shop aspects, and other miscellaneous tasks

2) Sourcing parts:

Arunima Basu will be responsible for this aspect of the project, as well as communication with the teaching staff for the grant.

3) Deadlines:

Bader AlKhawashki will be responsible for project documentation (written tasks), submitting the work, and keeping track of project deadlines, which includes reminders to the team and team members.

4) Absence procedure that your team can use to address future concerns like these if they arise.

Emails and messages will be sent multiple times prior to escalating the issue to the teaching team.

Deliverable 2:

For farmers/gardeners who own greenhouses, it's a robot that detects local humidity within a greenhouse and if it falls below a certain threshold, the robot extends a mechanism that sprays water mist and moves around the greenhouse to ensure proper humidification of the entire area. It also sprays nutrients on the soil every set number of hours using a retractable mechanism.

* Subject to Change.

(We opt in to be considered by the grant since this project improves the environment by taking care of plants in greenhouses.)

Deliverable 3:

Final Decision Matrix									
Scale:	5	1							
Cost	Cheapest	Expensive							
Feasibility	Simple to build	Hard to build							
Team enthusiasm	High	Low							
Aligned with Opportunity	Affirmative	Not at all							
Ease of use	Anyone can adopt	Elaborate instructions							

Our Greenhouse robot scored the best, and we chose it as our primary concept prototype.

Deliverables 4 (Imported our slides below)

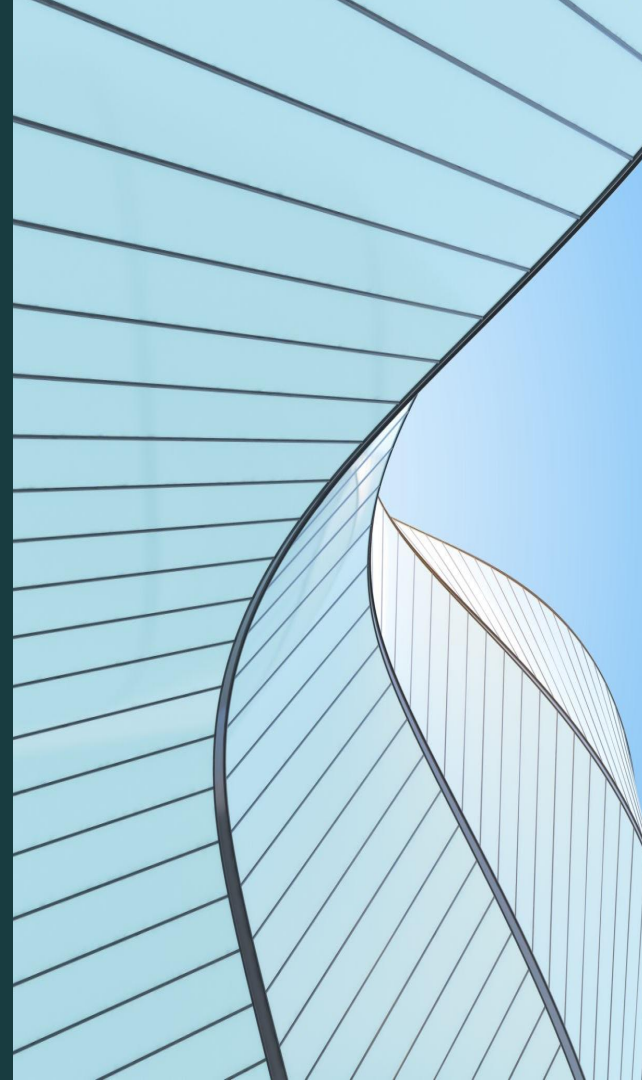
GreenBot

Greenhouse Robot : Mechatronics
Prototype project

ME 102B
Team #22

Team Members:

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Del. 4A

Farmers work long hours, but they cannot always be inside their greenhouses to constantly check humidity and adding nutrients to soils every set number of hours.

Even small changes in these conditions can affect plant health and crop yield.

Many existing automation systems are either too expensive or too complicated for everyday use, especially for small and medium sized farms. Because of this, farmers need a simple and affordable way to monitor and control their greenhouse environment even when they are not physically present.

Our GreenBot is designed to solve this problem by helping farmers easily manage greenhouse conditions remotely and automatically, reducing stress, saving time, and improving crop growth.

Del. 4B

GreenBot continuously monitors key greenhouse humidity conditions. When values drift outside target ranges, it automatically triggers the correct response (humidification).

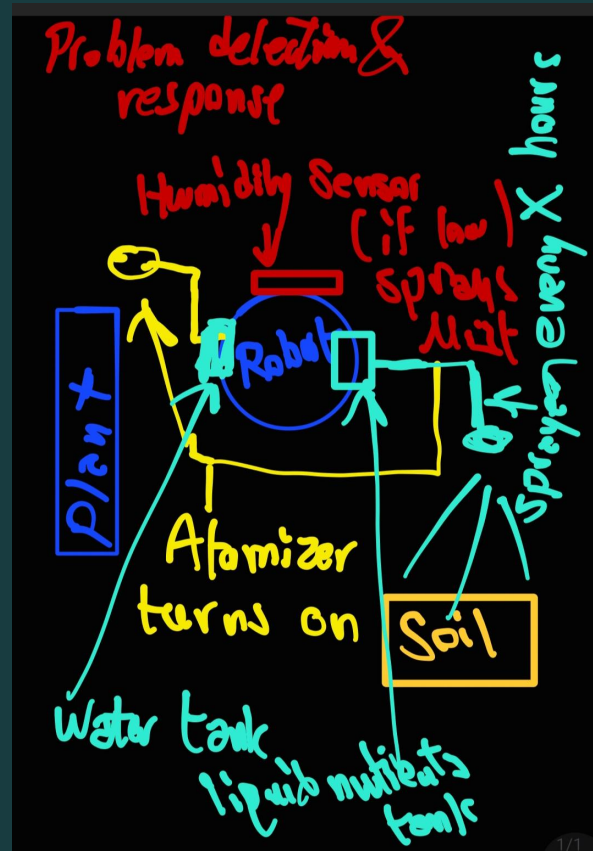


Figure 1: Deliverable 4.B Cartoon

* Degrees of Freedom: robot moves in any direction on the ground using motors. Sprayer can extend and retract (could be switched to static sprayer), humidity sensor stays static

** Subject to change

Del. 4C

Materials Exploration & Early Prototyping

We gathered basic materials to test key subsystems of our greenhouse robot, including a humidity sensor, small humidifier module, and a 4-legged robot. We also used a spray bottle with water to represent nutrient liquid delivery.

We play-acted the robot's behavior by manually moving the mockup through a marked greenhouse area while taking humidity readings at different locations and heights. When humidity dropped below a set threshold, we activated the humidifier to observe how quickly and how far moisture spread.

Key Observations

- Humidity varies across the greenhouse; it is not evenly distributed
- A stationary humidifier does not provide full coverage
- Robot movement improves humidity and liquid distribution
- Sensor placement near plant level is important

These tests confirmed that a mobile robot with automatic humidity control is a feasible and effective solution.

(Design subject to change.)*

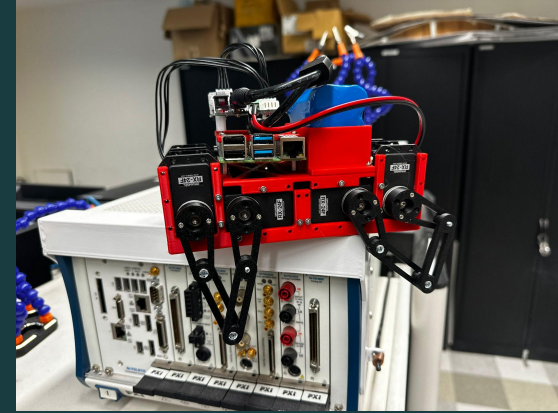


Figure 2: Early prototype picture

Del 4D

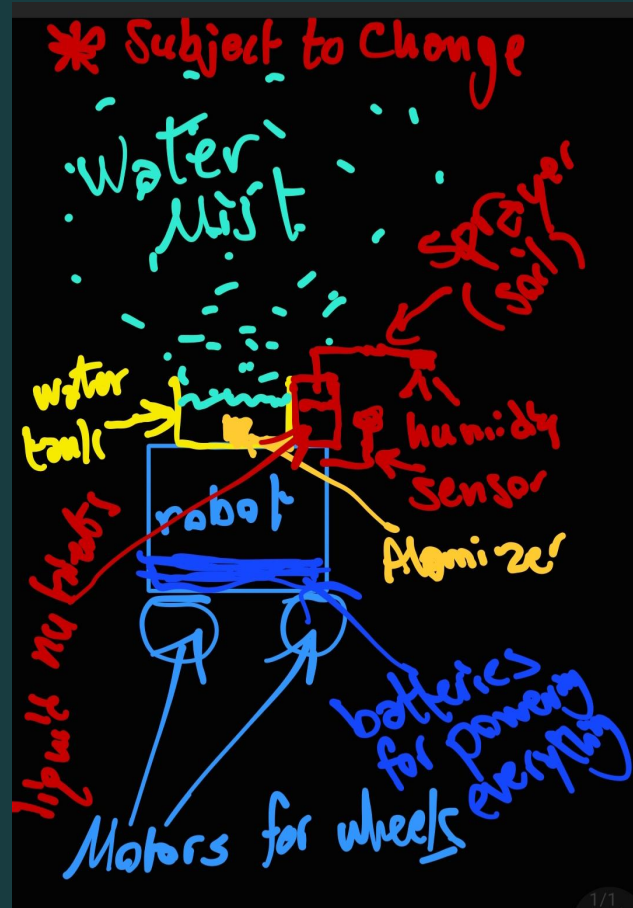


Figure 3: Deliverable 4.D Sketch *Subject to Change